

Problem Statement #22: Smart Parking Space Rental & License Validator

Use Case: UC-05 – Process Payment (Vehicle Exit)

Actor: Vehicle Owner

Preconditions:

- The vehicle has a valid recorded entry timestamp
- The vehicle's reservation, if any, has already been validated
- The parking fee tier table is configured and available in the system
- The vehicle is ready to exit the parking facility

Postconditions:

- The payment is successfully recorded against the vehicle's parking session
- A digital receipt is generated
- The vehicle's bay-occupancy/reservation record is closed
- The exit barrier is opened

Main Success Scenario:

1. Vehicle Owner arrives at the exit gate
2. Gate Camera System scans the vehicle's license plate
3. System retrieves the vehicle's entry timestamp and identifies the corresponding parking session
4. System invokes UC-04 Calculate Parking Fee to calculate the parking fee based on the elapsed parking duration and applicable tiered hourly rate
5. System displays the calculated fee to the Vehicle Owner
6. Vehicle Owner selects a payment method and confirms the payment
7. If the Vehicle Owner enters a valid discount/promo code, the system applies the discount (<<extend>> UC-06 Apply Discount / Promo)
8. System processes the payment

9. System records the successful payment and generates a digital receipt
10. System closes the bay-occupancy/reservation record
11. System opens the exit barrier
12. Use case ends successfully

Alternate Flow:

AF-01: Payment Declined

At Step 8:

1. System determines that the payment has failed
2. System displays a "Payment Failed" message to the Vehicle Owner
3. Vehicle Owner retries the payment using the same or another available payment method
4. If the payment succeeds, the system continues from Step 9 of the Main Success Scenario
5. If payment fails again after 2 attempts, the system flags the vehicle for Parking Warden review
6. The exit barrier remains closed until the issue is resolved